

Git Basics

Version Control System (VCS): A software tool that streamlines the process of managing file changes. It provides a structured approach to file management.

Git: A popular version control system renowned for its small data footprint, reliable handling, and deep feature set.

Repository (Repo): A collection of all versions of a folder over time.

In the Git data model, files are referred to as **blobs**, while folders are referred to as **trees**.

Commit: A snapshot of the state of a given folder.

Before changes to a repository can be committed, they must be **staged**, or added to the list of tracked files that will be included in the next commit.

Important Git Commands for Staging and Committing Changes

`git init` initializes an empty repository.

`git add .` adds all changes to staging.

`git commit -m "message"` adds a simple inline commit message without having to open an external text editor.

Commit data is stored in the form of **hashes**: strings of encoded, predictable length text that point to external data.

To visit a commit or branch in a repo, use the **checkout** command.

Branching and Merging

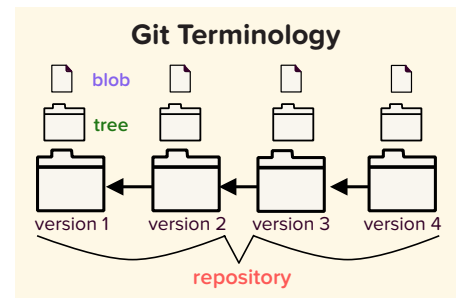
Branching allows commits to be stored in separate containers and developed in isolation. Branches have their own names and commits that belong to them.

A simple way to simultaneously create a new branch and check it out is `git checkout -b new-branch`.

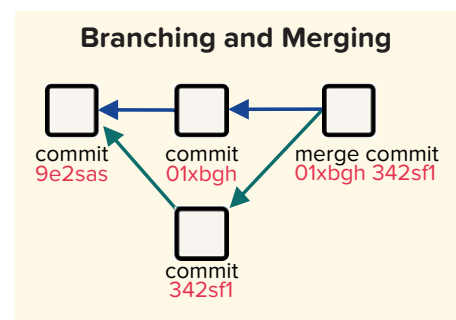
Merging attempts to combine two branches into a single merge commit.

If there are discrepancies between the two branches that Git cannot resolve, it throws a **merge conflict**.

Despite its inner simplicity, Git's command-line interface (CLI) is notoriously challenging and confusing. Using integrated development environment (IDE) plugins and graphical user interface (GUI) applications makes Git easier to work with.



Within Git, hashes are 40 characters long, but they are usually presented to the user with only their first six characters.



A **rebase** moves the contents of one branch into another branch without creating a new merge commit. It's an alternative to merging in situations where you want to avoid cluttering the commit history.

Changes can be **stashed** using the stash command, which saves changes to local memory.

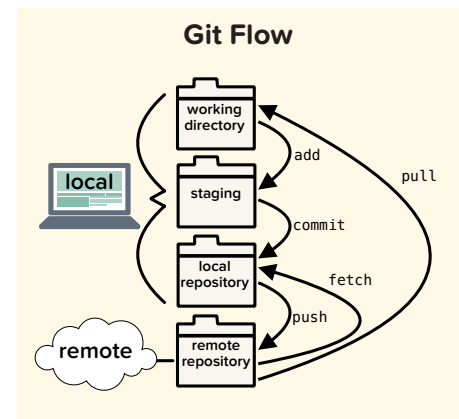
Remote Git and GitHub

Remote repositories: Git repos that are stored somewhere other than the user's working directory, usually on an internet server. They allow multiple people to coordinate their version control.

GitHub is the most popular hosting service for remote Git repositories.

The four most important remote commands are:

- **git pull**, which brings the remote's latest commits directly into the local working directory.
- **git push**, which moves new commits from the local repository into the remote.
- **git clone <url>**, which downloads a copy of a remote repository to your local machine.
- **git fetch**, which copies the new commits on the remote into the local repository (but not into the working directory).



Forks, Pull Requests, and Workflows

Forking creates a personal copy of a remote repository on your GitHub account.

A **pull request (PR)** asks the owner of the original repo for permission to incorporate your changes into their repo.

PRs can also be used within a repository via **protected branches**, which require a PR to be approved before they can be merged into.

Issues: Threaded messages on a repo's GitHub page that describe a problem, feature request, or idea.

Protected branches and PRs are a great way for teams to effectively collaborate on important projects.